R functions:

1. autoplot( ) ⇓ forecast pkg (fpp2) or ggplot( ) ⇓ ggplot2 (tidyverse)

input: ts( ) object

input: for line charts, df with a x & y

output: we look to see if the ts is: (a) stationary? (do we have a trend / seasonali
(b) do we have missing data?

2. we use the successive kpss. tests through the ndiffs( ) (forecast) to confirm whether data is stationary [ndiffs = 0] or not [ndiffs > 0]

input: ts( ) object → must if you have a freq > 1

3. we look @ the acf of the (differenced, if ndiffs > 0) ts
   - → acf( ) ⇒ base R [starts @ lag 0, which always = 1)
   - → acf(, plot = F) %>% autoplot( ) } ignores lag 0
     - %>% ↓ dplyr
     - autoplot( ) ↓ forecast

similar for pacf

output: we look for

	acf	pacf
MA(q)	cuts off at lag q	ignore
AR(p)	exponential or sinsodial pattern	cuts of @ lag p
ARMA	exp or sinsodial pat.	

4. model using Arima( ) or auto.arima( ) ⟹ both come from forecast pkg

inputs:
- y → ts
- p, d, q
- P, D, Q

Recall $y_t = \delta + \phi_1 y_{t-1} + \dots + \phi_p y_{t-p}$ (AR(p))

$y_t = \mu + \varepsilon_t - \theta_1 \varepsilon_{t-1} - \dots - \theta_q \varepsilon_{t-q}$ (MA(q))

what we look for in output:
- summary → goodness of fit measures; estimated coefficients; pred. acc metrics
- checkresiduals → if p-value > 0.05 then we don't have enough evidence to suggest that residuals are correlated.

* input for both an object of type ARIMA

5. if happy, forecast( ) to predict h time periods ahead